12 Saving data with Entity Framework Core

This chapter covers

- Understanding what Entity Framework Core is and why you should use it
- Adding Entity Framework Core to an ASP.NET Core application
- Building a data model and using it to create a database
- Querying, creating, and updating data with Entity Framework Core

Most applications that you'll build with ASP.NET Core require storing and loading some kind of data. Even the examples so far in this book have assumed that you have some sort of data store—storing exchange rates, user shopping carts, or the locations of physical stores. I've glossed over this topic for the most part, but typically you'll store this data in a database.

Working with databases can be a rather cumbersome process. You have to manage connections to the database, translate data from your application to a format the database can understand, and handle a plethora of other subtle problems. You can manage this complexity in a variety of ways, but I'm going to focus on using a library built for modern .NET: Entity Framework Core (EF Core). EF Core is a library that lets you quickly and easily build database access code for your ASP.NET Core applications. It's modeled on the popular Entity Framework 6.x library, but it has significant changes that make it stand alone in its own right as more than an upgrade.

The aim of this chapter is to provide a quick overview of EF Core and show how you can use it in your applications to query and save to a database quickly. You'll learn enough to connect your app to a database and manage